

Structured Step-Size Policies for Scalable Worst-Case Optimization Design

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How to optimize an optimization algorithm?

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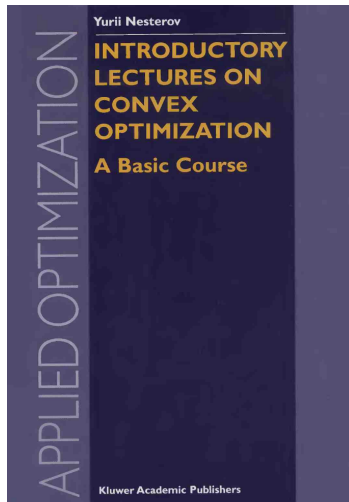
Ultimate Goal

$$\min_{\alpha \in \mathbb{R}^N} \sup_{f \in \mathcal{F}, x_n = \mathcal{M}_\alpha(f, x_1)} \mathcal{P}(f, x_1, \dots, x_N)$$

Option 1: Have Yurii Nesterov in your lab



Figure: Yurii Nesterov



Option 2: Performance Estimation Problems (PEP)¹

Finding the **worst-case** of $\mathcal{M}_\alpha$ for $\left\{ \begin{array}{l} N : \text{Number of Iterations} \\ \mathcal{P} : \text{Performance Metric} \\ \mathcal{F} : \text{Class of functions} \\ \mathcal{I} : \text{Initial Conditions} \end{array} \right.$ amounts to solve

PEP: In general

$$\begin{aligned} \sup_{f, x_1, \dots, x_N, x_*} \quad & \mathcal{P}(f, x_1, \dots, x_N, x_*) \\ \text{s.t.} \quad & f \in \mathcal{F}, \\ & x_* \text{ is optimal for } f, \\ & x_2, \dots, x_N \sim \mathcal{M}_\alpha \\ & \mathcal{I}(f, x_1, \dots, x_N, x_*) \leq R \end{aligned}$$

¹Adrien B Taylor, Julien M Hendrickx, and François Glineur. "Smooth strongly convex interpolation and exact worst-case performance of first-order methods". In: *Mathematical Programming* 161 (2017), pp. 307–345.

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PEP: Applied to Gradient Descent

$$\sup_{f, x_1, \dots, x_N, x_*} f(x_N) - f(x^*)$$

subject to $f \in \mathcal{F}$,

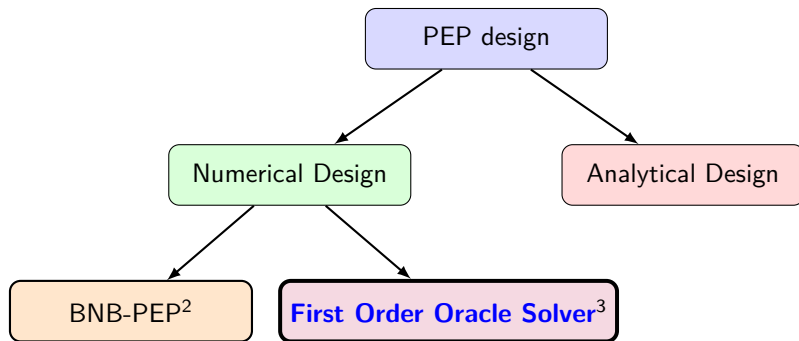
$$\nabla f(x_*) = 0,$$

$$x_{n+1} = x_n - \alpha_n \nabla f(x_n)$$

$$\|x_1 - x_*\| \leq R$$

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PEP Design Tree



²Shuvomoy Das Gupta, Bart PG Van Parys, and Ernest K Ryu. "Branch-and-bound performance estimation programming: a unified methodology for constructing optimal optimization methods". In: *Mathematical Programming* 204 (2024), pp. 567–639.

³Yassine Kamri, Julien M. Hendrickx, and François Glineur. *Numerical Design of Optimized First-Order Algorithms*. 2025. arXiv: 2507.20773 [math.OC]. URL: <https://arxiv.org/abs/2507.20773>.

Outline

1 Background: Performance Estimation Problems (PEP)

- Interpolation Constraints
- PEP as an SDP

2 Numerical Design of Methods

- First-order information $\nabla_{\alpha} W$
- Second-order information $\nabla_{\alpha}^2 W$

3 Parameterized Step-sizes Mappings

- Goal
- First and second information w.r.t. parameters; $\nabla_{\omega} W(\alpha(\omega)), \nabla_{\omega}^2 W(\alpha(\omega))$
- Numerical Experiments

4 Bonus: Anytime Design of Methods

- Multiple Horizons Design Problem

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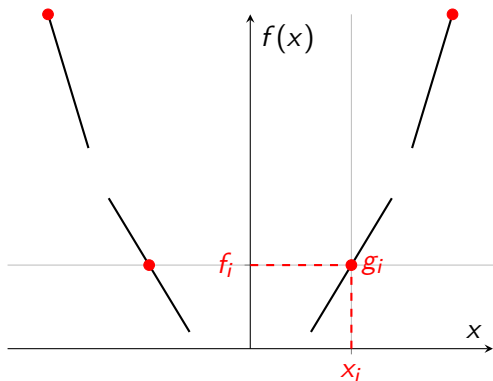
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Interpolation Constraints (IC)

Definition

Given a class of functions $\mathcal{F}$, the $\{(x_i, g_i, f_i)\}_{i \in I}$ is $\mathcal{F}$ -interpolable iff there exists $f \in \mathcal{F}$ such that $f_i = f(x_i)$, $g_i = \nabla f(x_i)$, $\forall i \in I$.



Interpolation Constraints

Convex:

$$f_j \geq f_i + \langle g_i, x_j - x_i \rangle, \quad \forall i, j$$

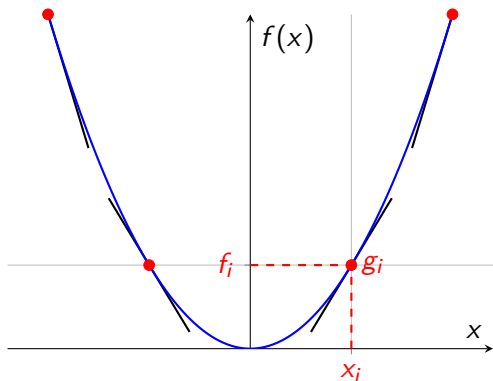
Strongly-convex:

$$f_j \geq f_i + \langle g_i, x_j - x_i \rangle + \frac{\mu}{2} \|x_i - x_j\|^2, \quad \forall i, j$$

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PEP as Semi-Definite Program

Let be

$$\mathbf{G} = \left[\begin{array}{c|c} \mathbf{x}_i^T \mathbf{x}_j & \mathbf{x}_i^T \mathbf{g}_j \\ \hline \mathbf{g}_i^T \mathbf{x}_j & \mathbf{g}_i^T \mathbf{g}_j \end{array} \right] \in \mathbb{R}^{2N \times 2N}$$

and

$$\mathbf{F} = [f_1 \dots f_n]^T$$

Reformulating Constraints as LMI

$$\begin{aligned} f_j &\geq f_i + \langle \mathbf{g}_i, \mathbf{x}_j - \mathbf{x}_i \rangle \\ \iff 0 &\geq f_i - f_j + \langle \mathbf{g}_i, \mathbf{x}_j \rangle - \langle \mathbf{g}_i, \mathbf{x}_i \rangle \\ \iff 0 &\geq \mathbf{F}^T (\mathbf{u}_i - \mathbf{u}_j) + \text{Tr}(\mathbf{A}_{ij} \mathbf{G}) \end{aligned}$$

PEP as Semi-Definite Program

Let be

$$\mathbf{G} = \left[\begin{array}{c|c} x_i^T x_j & x_i^T g_j \\ \hline g_i^T x_j & g_i^T g_j \end{array} \right] \in \mathbb{R}^{2N \times 2N}$$

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BUT

$$\mathbf{x}_n = \mathbf{x}_{n-1} - \alpha_{n-1} \mathbf{g}_{n-1} = \mathbf{x}_1 - \sum_{k=1}^{n-1} \alpha_k \mathbf{g}_k = \mathbf{x}_1 - [\alpha_1 \dots \alpha_{k-1} \ 0 \ 1] \begin{bmatrix} g_1 \\ \vdots \\ g_n \\ 1 \end{bmatrix}$$

PEP as Semi-Definite Program

Let be

$$G = \begin{bmatrix} x_1^T x_1 & x_1^T g_j \\ g_i^T x_1 & g_i^T g_j \end{bmatrix} \in \mathbb{R}^{N+1 \times N+1}$$

and

$$F = [f_1 \dots f_n]^T$$

Reformulating Constraints as LMI

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$$\begin{aligned} W^P(\alpha) &= \sup_{G \in \mathbb{S}^N, F \in \mathbb{R}^N} b^T F + \text{Tr}(C G) && \text{(Linear + Bilinear terms in } \mathcal{P}(\cdot)) \\ \text{s.t. } F^T (\mu_i u_i - \mu_j u_j) &\geq \text{Tr}(A_{i,j}(\alpha) G), \quad \forall i, j \in I && \text{(Interpolation Constraints)} \\ \text{Tr}(A_R G) - R^2 &\leq 0, && \text{(Initial Conditions)} \\ G &\succeq 0 && \text{(Norm is well-defined)} \end{aligned}$$

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Computing $\nabla_{\alpha} W$ using the Dual PEP

$$W^P(\alpha) = \sup_{G \in \mathbb{S}^N, F \in \mathbb{R}^N} b^T F + \text{Tr}(C G)$$

$$\text{s.t. } F^T(\mu_i u_i - \mu_j u_j) \geq \text{Tr}(\mathbf{A}_{i,j}(\alpha) G)$$

$$\text{Tr}(A_R G) - R^2 \leq 0$$

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Primal

⁴Yassine Kamri, Julien M. Hendrickx, and François Glineur. *Numerical Design of Optimized First-Order Algorithms*. 2025. arXiv: 2507.20773 [math.OC]. URL: <https://arxiv.org/abs/2507.20773>.

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Primal

$$\begin{aligned} W^D(\alpha) &= \inf_{\tau, \lambda_{i,j}} \tau R^2 \\ \text{s.t. } \tau A_R - C + \sum_{i,j \in I} \lambda_{i,j} \mathbf{A}_{i,j}(\alpha) &\succeq 0 \\ b - \sum_{i,j} \lambda_{i,j}(\mu_j u_j - \mu_i u_i) &= 0 \\ \lambda_{i,j} &\geq 0 \end{aligned}$$

Dual

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$$G \succeq 0$$

Primal

Strong duality

$$W^P(\alpha) = W^D(\alpha)$$

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$$\text{s.t. } \tau A_R - C + \sum_{i,j \in I} \lambda_{i,j} \mathbf{A}_{i,j}(\alpha) \succeq 0$$

$$b - \sum_{i,j} \lambda_{i,j} (\mu_j u_j - \mu_i u_i) = 0$$

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$$\lambda_{i,j} \geq 0$$

Dual

Theorem (Subgradient of W w.r.t. α by⁴)

$$\frac{\partial W}{\partial \alpha_n}(\alpha) = \text{Tr} \left(-G \left[\sum_{i,j \in I} \lambda_{i,j} \frac{\partial \mathbf{A}_{i,j}(\alpha)}{\partial \alpha_n} \right] \right)$$

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Taking a step-back

- Design of a method $\mathcal{M}_\alpha$

$$\min_{\alpha \in \mathbb{R}^N} \sup_{f, x_{n+1} = \mathcal{M}_\alpha(x, f)} \underbrace{\mathcal{P}(f, x_1, \dots, x_N)}_{=: W(\alpha)} \quad (\text{DP})$$

- Minimizing W w.r.t. α

$$\alpha^{(t+1)} \leftarrow \alpha^{(t)} - \beta \nabla_\alpha W(\alpha^{(t)})$$

Taking a step-back

- Design of a method $\mathcal{M}_\alpha$

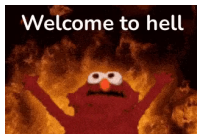
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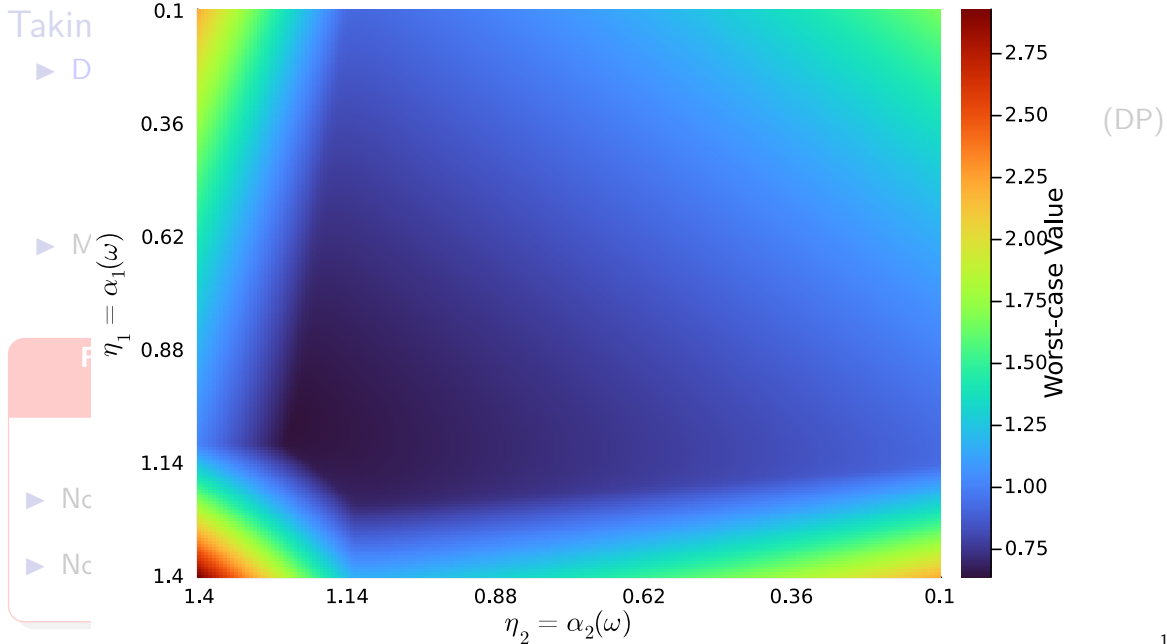
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Fundamental Limitations of the Design Problem

- Non-convex
- Non-smooth





Taking a step-back

- Design of a method $\mathcal{M}_\alpha$

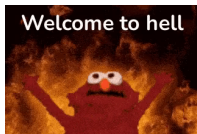
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Limitations of the Literature

- Use the first-order information only
- Dependency on N
 - α_N^* cannot be used for $N + 1$ iterations
 - Size of the search space grows linearly with N

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Second order information w.r.t. α

Why is it useful?

First-order Methods Linear convergence rate.

Second-order Methods Quadratic convergence rate (near the optimum) :).

Theorem (Expression for $\nabla_{\alpha}^2 W$)

The (n, p) -th entry of $\nabla_{\alpha}^2 W$ is given by

$$\frac{\partial^2 W}{\partial \alpha_n \partial \alpha_p}(\alpha) = \text{Tr} \left(-\mathbf{G} \left[\sum_{i,j \in I} \lambda_{i,j} \frac{\partial^2 A_{i,j}(\alpha)}{\partial \alpha_n \partial \alpha_p} \right] \right)$$

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Parameterized Step-sizes

Free step-sizes

$$\min_{\alpha \in \mathbb{R}^N} W(\alpha) = \min_{\alpha \in \mathbb{R}^N} \sup_{f \in \mathcal{F}, x_n = \mathcal{M}_\alpha(f, x_1)} \mathcal{P}(f, x_1, \dots, x_N)$$

$\leadsto$ the dimension of the search space grows with N .

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Parameterized step-sizes

$$\min_{\omega \in \mathbb{R}^K} W(\alpha(\omega)) = \min_{\omega \in \mathbb{R}^K} \sup_{f \in \mathcal{F}, x_n = \mathcal{M}_{\alpha(\omega)}(f, x_1)} \mathcal{P}(f, x_1, \dots, x_N)$$

Parameterized Step-sizes

Free step-sizes

$$\min_{\alpha \in \mathbb{R}^N} W(\alpha) = \min_{\alpha \in \mathbb{R}^N} \sup_{f \in \mathcal{F}, x_n = \mathcal{M}_\alpha(f, x_1)} \mathcal{P}(f, x_1, \dots, x_N)$$

$\leadsto$ the dimension of the search space grows with N .

Parameterized step-sizes

$$\min_{\omega \in \mathbb{R}^\kappa} W(\alpha(\omega)) = \min_{\omega \in \mathbb{R}^\kappa} \sup_{f \in \mathcal{F}, x_n = \mathcal{M}_{\alpha(\omega)}(f, x_1)} \mathcal{P}(f, x_1, \dots, x_N)$$

where $\alpha : \mathbb{R}^\kappa \longrightarrow \mathbb{R}^N$ can be, for example,

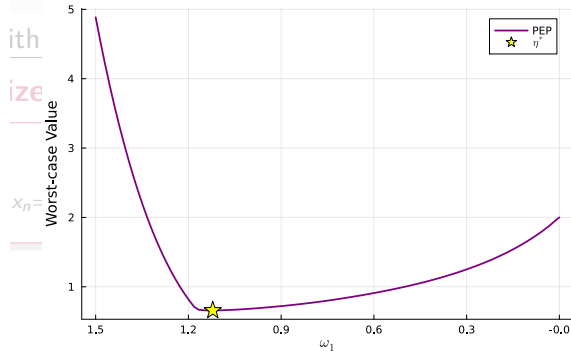
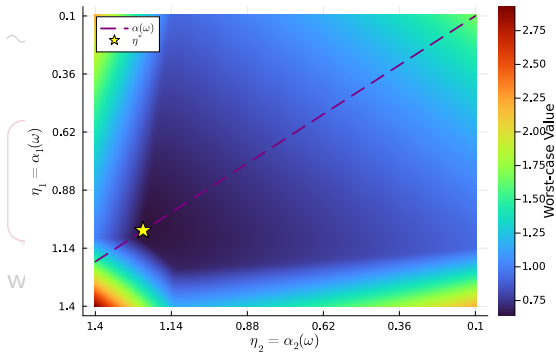
$$\alpha(\omega) \in \left\{ \frac{\omega_1}{n^{\omega_2} + \omega_3}, \quad \frac{\omega_1}{\log(n)^{\omega_2} + \omega_3}, \quad \sum_{i=0}^d \omega_i n^i, \dots \right\}$$

where $\kappa \ll N$.

Parameterized Step-sizes

Free step-sizes

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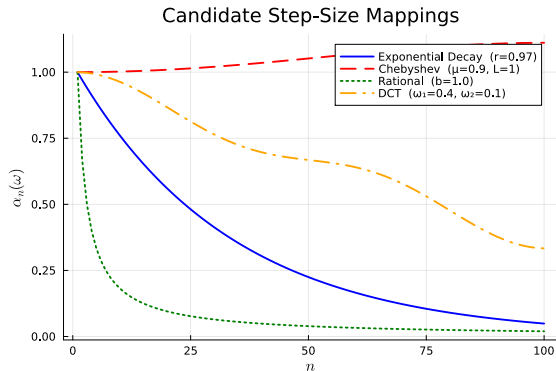
$$\alpha(\omega) \in \left\{ \overline{n^{\omega_2} + \omega_3}, \overline{\log(n)^{\omega_2} + \omega_3}, \sum_{i=0}^{\omega_1} \omega_i, \dots \right\}$$

where $\kappa \ll N$.

Some Candidates for the Step-Sizes Mappings

Close form mappings

$$\begin{aligned}\alpha_n(\omega) &:= \sum_{i=1}^k \omega_i^{(1)} e^{-\omega_i^{(2)} n} + c, & (\text{Exponential Decay}) \\ \alpha_n(\omega) &:= \sum_{k=0}^K \omega_k T_k(n), & (\text{Chebyshev Polynomials}) \\ \alpha_n(\omega) &:= \frac{\omega_0 + \omega_1 n}{1 + \omega_1 n + \omega_2 n^2}, & (\text{Rationals}) \\ \alpha_n(\omega) &:= \sum_{k=0}^K \omega_k \cos\left(\frac{\pi k(n-1)}{N-1}\right), & (\text{Discrete Cosine Transform}) \\ & \vdots\end{aligned}$$



Gradient and Hessian w.r.t. parameters ω

Theorem (Expression for $\nabla_{\omega} W(\alpha(\omega))$ and $\nabla_{\omega}^2 W(\alpha(\omega))$)

Given a parameterized step-sizes functions $\alpha : \mathbb{R}^{\kappa} \longrightarrow \mathbb{R}^N$ whose parameters are $\omega \in \mathbb{R}^{\kappa}$, we have

$$\frac{\partial W}{\partial \omega_k}(\alpha(\omega)) = \sum_{n=1}^N \frac{\partial \alpha_n(\omega)}{\partial \omega_k} \frac{\partial W(\alpha(\omega))}{\partial \alpha_n}$$

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$$\frac{\partial W}{\partial \omega_k}(\alpha(\omega)) = \sum_{n=1}^N \frac{\partial \alpha_n(\omega)}{\partial \omega_k} \frac{\partial W(\alpha(\omega))}{\partial \alpha_n}$$

and

$$\frac{\partial^2 W}{\partial \omega_k \partial \omega_s}(\alpha(\omega)) = \sum_{n=1}^N \frac{\partial^2 \alpha_n(\omega)}{\partial \omega_s \partial \omega_k} \frac{\partial W(\alpha(\omega))}{\partial \alpha_n} + \frac{\partial \alpha_n}{\partial \omega_k} \sum_{m=1}^N \frac{\partial \alpha_m}{\partial \omega_s} \frac{\partial^2 W(\alpha(\omega))}{\partial \alpha_n \partial \alpha_m}$$

for $k, s \in [\kappa]$.

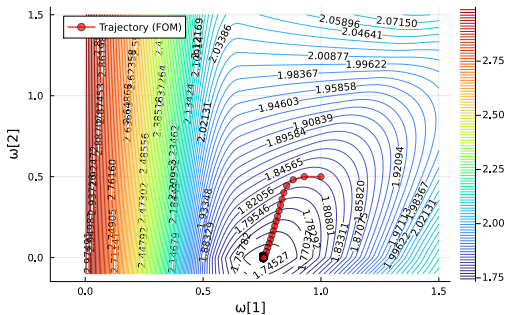
Using first and second-order information

First-Order Method (FOM)

Gradient Descent

$$\omega^{t+1} = \omega^{(t)} - \beta \nabla_{\omega} W(\alpha(\omega^{(t)}))$$

Landscape of the PEP (ω -space)



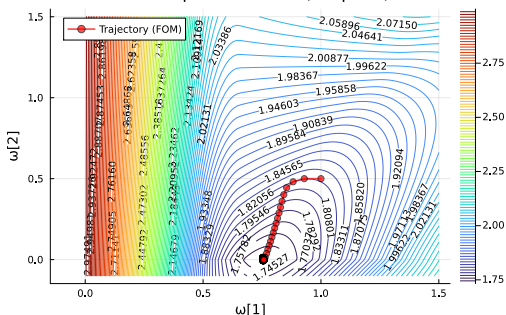
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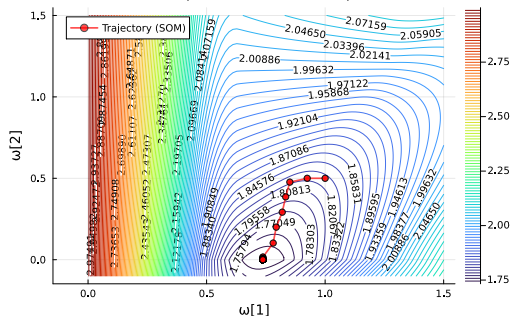


Second-Order Method (SOM)

Trust-Region Method

$$\arg \min_{\mathbf{d}} \frac{1}{2} \mathbf{d}^T \nabla^2 W(\alpha(\omega^{(t)})) \mathbf{d} + \nabla W(\alpha(\omega^{(t)}))^T \mathbf{d}$$

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- PEP as an SDP

2 Numerical Design of Methods

- First-order information $\nabla_{\alpha} W$
- Second-order information $\nabla_{\alpha}^2 W$

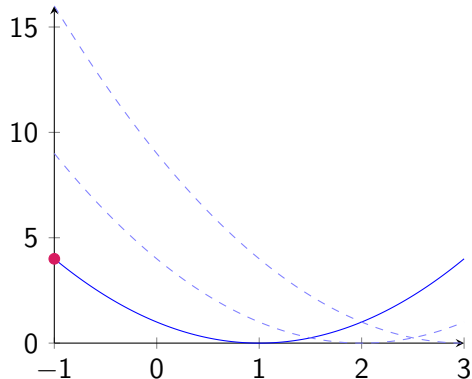
3 Parameterized Step-sizes Mappings

- Goal
- First and second information w.r.t. parameters; $\nabla_{\omega} W(\alpha(\omega)), \nabla_{\omega}^2 W(\alpha(\omega))$
- Numerical Experiments

4 Bonus: Anytime Design of Methods

- Multiple Horizons Design Problem

Online Optimization



Receive: $f_1, \dots, f_N$ sequentially

Setting:

A1 : f_t is L -smooth

A2 : Feasible set bound by diameter D

Online Gradient Descent

$$\mathcal{M}_{\alpha(\omega)} \quad \begin{array}{l} x_{n+1} = x_n - \alpha_n(\omega) \nabla f_n(x_n) \\ n = 1 \end{array}$$

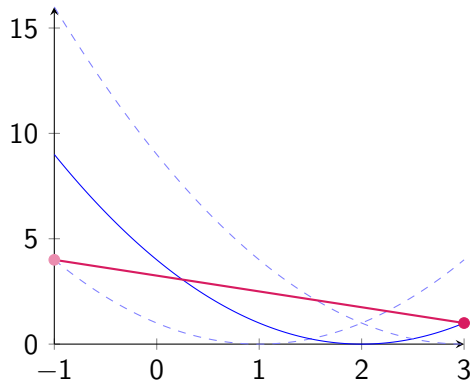
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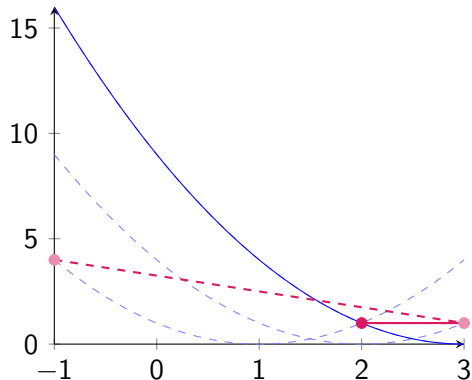


Online Gradient Descent

$$\mathcal{M}_{\alpha(\omega)} \quad x_{n+1} = x_n - \alpha_n(\omega) \nabla f_n(x_n)$$

$n = 2$

Online Optimization



Receive: $f_1, \dots, f_N$ sequentially

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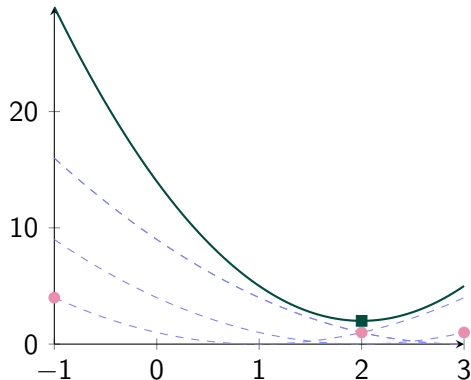
A2 : Feasible set bound by diameter D

Online Gradient Descent

$$\mathcal{M}_{\alpha(\omega)} \quad x_{n+1} = x_n - \alpha_n(\omega) \nabla f_n(x_n)$$

$n = 3$

Online Optimization



Receive: $f_1, \dots, f_N$ sequentially

Setting:

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Online Gradient Descent

$$\mathcal{M}_{\alpha(\omega)} \quad \begin{aligned} x_{n+1} &= x_n - \alpha_n(\omega) \nabla f_n(x_n) \\ n &\in \{1, 2, 3\} \end{aligned}$$

Performance metric $\mathcal{P}$:

$$\text{Regret} = \underbrace{\sum_{n=1}^N f_n(x_n)}_{\text{Online}} - \underbrace{\min_x \sum_{n=1}^N f_n(x_*)}_{\text{Offline}}$$

Inexact Gradient Descent

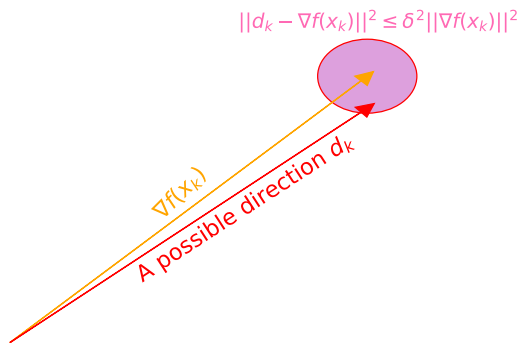


Figure: by Pierre Vernimmen.

Setting

A1 f is L -smooth

A2 Inexact Gradient oracle d such that

$$\exists \delta \in [0, 1), \quad \|\nabla f(x) - d(x)\| \leq \delta \|\nabla f(x)\|, \quad \forall x$$

Inexact Gradient Descent Method

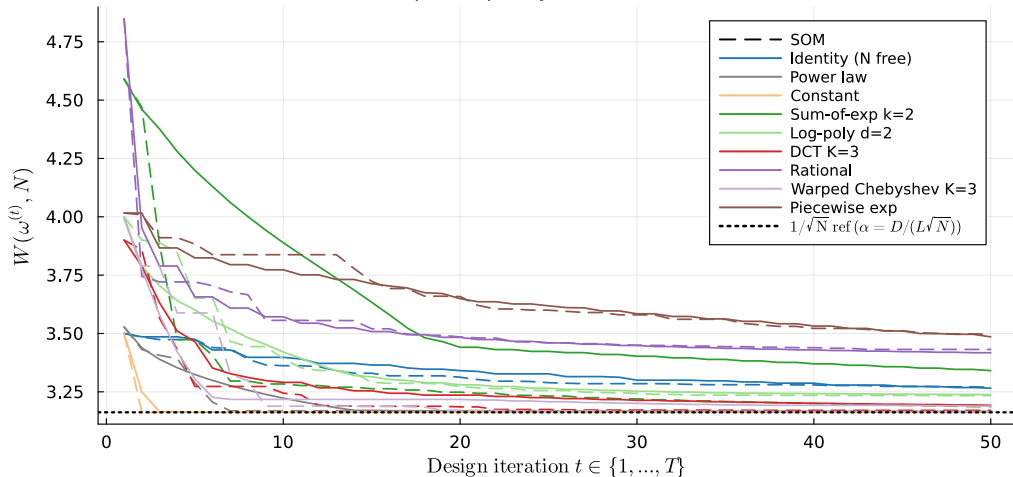
$$\mathcal{M}_{\alpha(\omega)} \quad x_{n+1} = x_n - \alpha_n(\omega) d(x_n)$$

Performance Metric $\mathcal{P}$

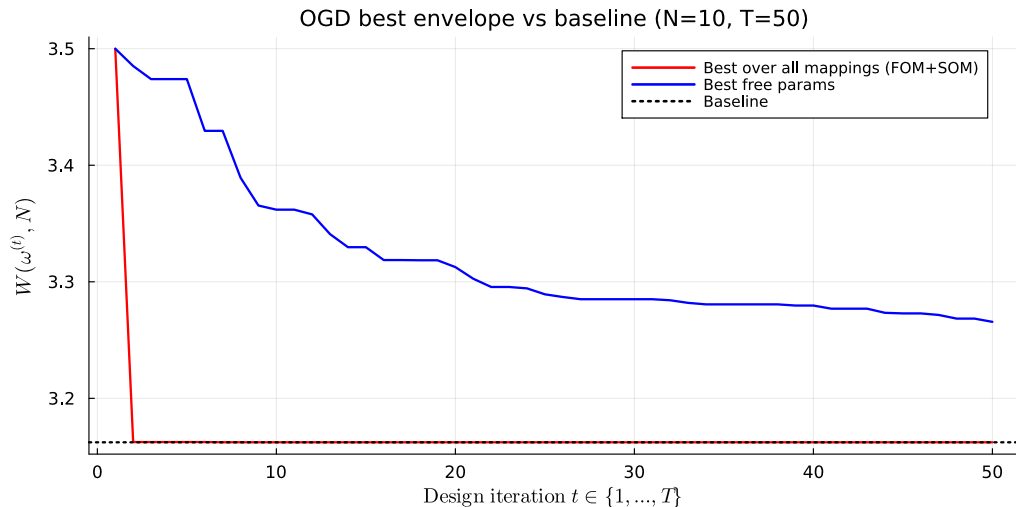
$$\text{Best gradient} \quad \min_{k=1, \dots, N} \|g_k\|$$

Numerical Design of Online Gradient Descent

OGD step-size policy search (N=10, T=50)

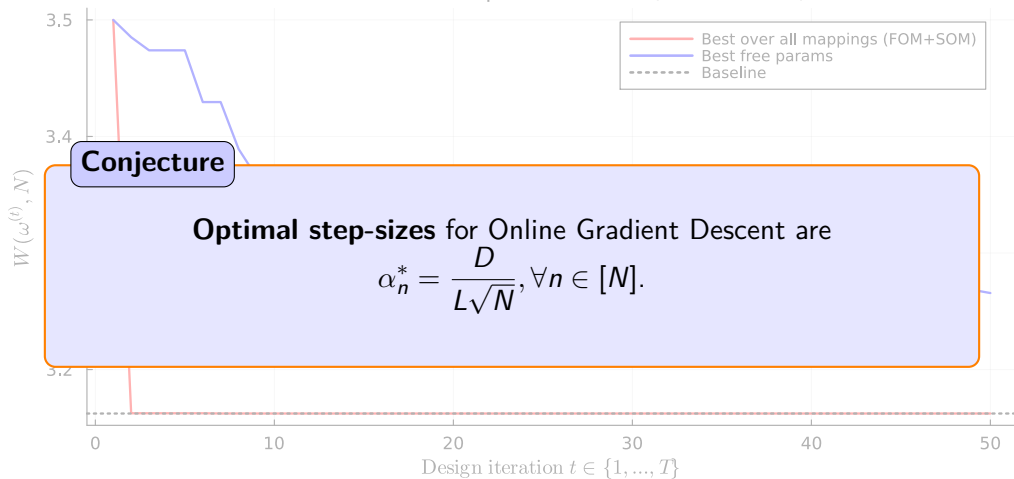


Numerical Design of Online Gradient Descent

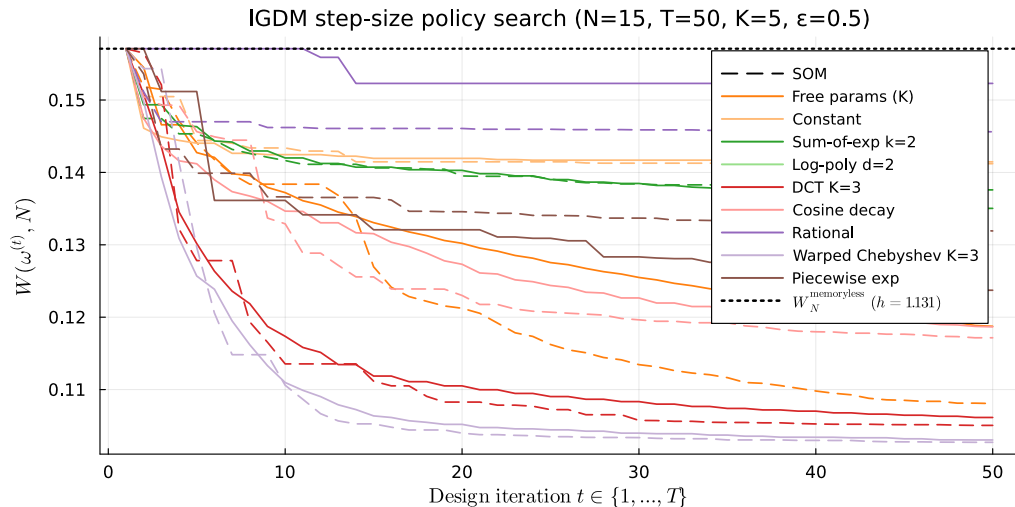


Numerical Design of Online Gradient Descent

OGD best envelope vs baseline (N=10, T=50)

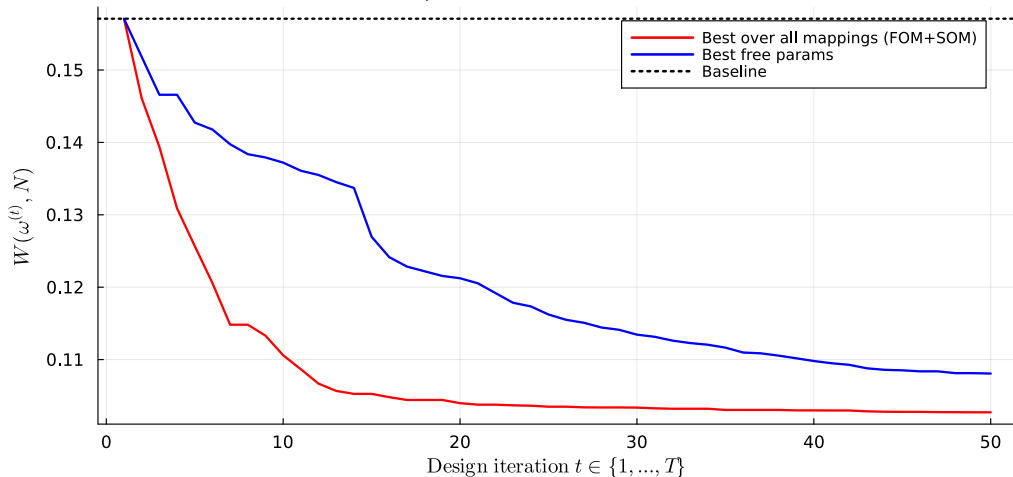


Numerical Design of Inexact Gradient Descent with Memory



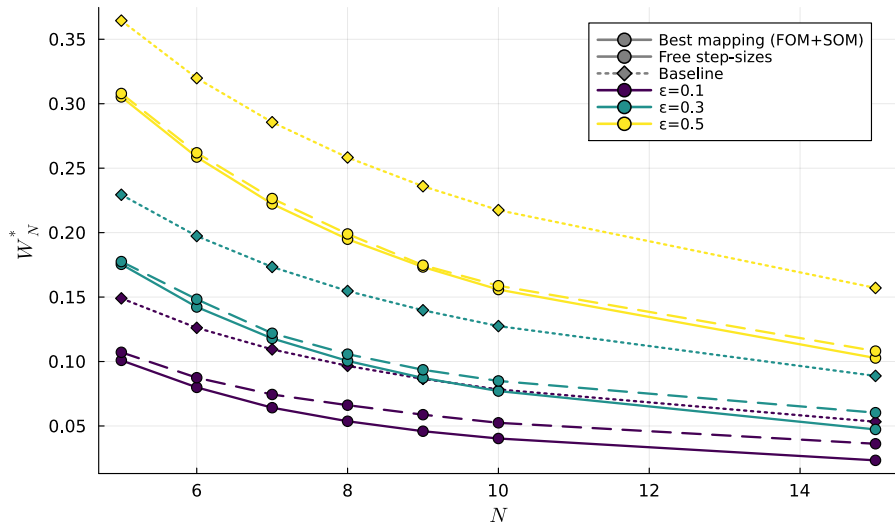
Numerical Design of Inexact Gradient Descent with Memory

IGDM best envelope vs baseline (N=15, T=50, K=5, $\varepsilon=0.5$)



Numerical Design of Inexact Gradient Descent with Memory

IGDM best performance vs N ($T=50$, $K=5$)



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So far we have solved,

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- 1 Take **another horizon** $K \neq N$. Does ω_N^* , still perform well for $W(\alpha(\omega_N^*), K)$?
- 2 How can we leverage this framework **to obtain good step-sizes for any N** ?

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$$\omega \mapsto \sum_{N \in \mathcal{H}} \frac{W(\alpha(\omega), N)}{\min_{\omega_N} W(\alpha(\omega_N), N)}$$

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Goal

$$\min_{\omega} \sigma(\mathcal{H}, \omega) := \frac{1}{|\mathcal{H}|} \sum_{N \in \mathcal{H}} \underbrace{\frac{W(\alpha(\omega), N)}{\min_{\omega_N} W(\alpha(\omega_N), N)}}_{=: f_N(\omega)}$$

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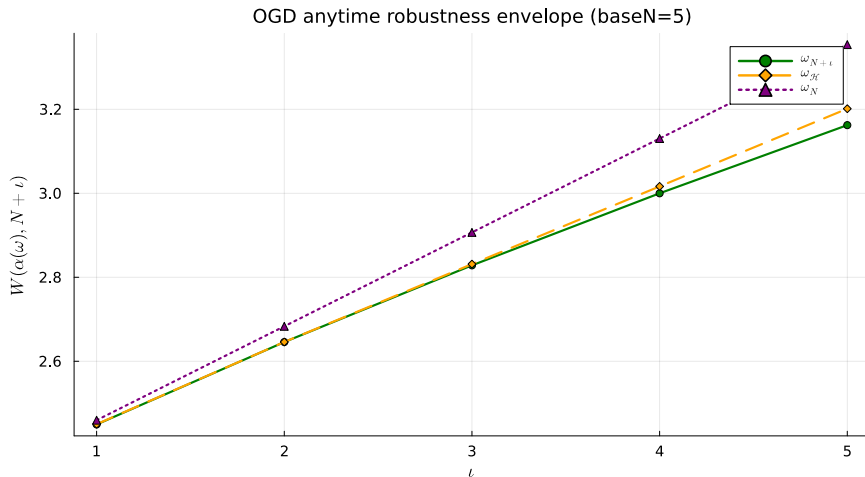
$$\nabla_{\omega} \sigma(\mathcal{H}, \omega) = \frac{1}{|\mathcal{H}|} \sum_{N \in \mathcal{H}} \nabla_{\omega} f_N(\omega) \text{ and } \nabla_{\omega}^2 \sigma(\mathcal{H}, \omega) = \frac{1}{|\mathcal{H}|} \sum_{N \in \mathcal{H}} \nabla_{\omega}^2 f_N(\omega)$$

$\implies$ Computation of σ , $\nabla_{\omega} \sigma$ and $\nabla_{\omega}^2 \sigma$ is parallelizable over $\mathcal{H}$.

Robustness of step-sizes to additional iterations

How does ω_N perform for $N + \iota$ iterations?

$\omega_N := \arg \min W(\alpha(\omega), N)$, $\omega_{N+\iota} := \arg \min W(\alpha(\omega), N + \iota)$, $\omega_{\mathcal{H}} := \arg \min \sigma(\alpha(\omega), \mathcal{H})$



Conclusion

Main Idea

$$\min_{\omega \in \mathbb{R}^{\kappa}} W(\alpha(\omega)) = \min_{\omega \in \mathbb{R}^{\kappa}} \sup_{f \in \mathcal{F}, x_n = \mathcal{M}_{\alpha(\omega)}(f, x_1)} \mathcal{P}(f, x_1, \dots, x_N)$$

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Inexact Gradient Descent

Better improvements than
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Future works

- ▶ Obtaining convergence guarantees for the design problem using Clarke's sub-differential
- ▶ Anytime; transferring the knowledge gained from $\mathcal{H}_{small}$ to another set of horizons $\mathcal{H}_{big}$

Thanks for your attention! Questions?

`erwan.meunier@uclouvain.be`

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